

Design the Decision Process

Project: Helpful Classroom Sorter

1. Design the Decision Process

Teach the learning cycle

Primary target: Design a decision process that covers valid, invalid and no-match requests

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose Inputs and Outputs. Give an example and a mistake that the idea helps you avoid.

Look for: Turn the project outcome into precise input and output requirements

Reference example, not the only acceptable wording: Time required must be no more than available time; when quiet is required the activity must be quiet. Choose the first compatible activity in B1 to B4 order. No means quiet is not required, not forbidden, so quiet B1 is permitted if 10 minutes fits.

Humans Set Boundaries

Without opening the old lesson, explain one key idea from Humans Set Boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a system's capability, the correctness of its output and permission to take an action

Reference example, not the only acceptable wording: The button establishes a capability. The fact check establishes factual support for A at that time. The approval permits the specified classroom-board action within its limits. Public posting is outside that permission even though the button exists and A is factually checked.

Generate, Retrieve, or Calculate?

Without opening the old lesson, explain one key idea from Generate, Retrieve, or Calculate?. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish retrieving a record, calculating from inputs and generating wording

Reference example, not the only acceptable wording: Room: retrieval, checked against the right source N2 and event. Remaining places: calculation, checked as $6 \times 4 - 3 = 21$ with correct inputs. Title: generation, checked for the requested style and unsupported factual details. An application can combine them; the labels describe operations rather than exclusive types of application.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Rule: if paper then P, otherwise M. Inputs are paper, metal, glass and unknown. Repair the rule for paper and metal trays without inventing a glass tray. Trace all four inputs.

Concept to explain

Rules or learned patterns must connect inputs to outputs transparently.

Term · Meaning

Term: decision flow

Meaning: ordered path connecting inputs to outputs

Term: baseline

Meaning: simple method used for comparison

Term: branch

Meaning: path chosen by a condition

Term: explanation

Meaning: traceable reason for an output

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Keep Week 30's requirements separate from the algorithm that implements them. The manual baseline marked all rows; this design may stop at the first fit while preserving the required output. Check more than final labels: pupils must explain both conditions, validation and progress through the finite catalogue. This is an explicit-rule recommender, not a model fitted from examples.

Readiness answer

Blank time requires clarification because an input is missing. Zero is valid within 0 to 30, but no activity in the catalogue fits it, so the correct result is no match.

The unchanged B1 to B4 catalogue and Week 30 requirements

Four candidate cards and a marker for the current row

Paper for a process drawing and a trace with candidate, checks and next step

End goal for the pictorial case

By the end, I can design a complete decision flow with validation, search and stopping.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain the mechanism

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

INPUT approved input fields

PROCESS smallest rule or learned decision flow with visible branches

OUTPUT suggested bin, not sure, and reason

CORE IDEA Rules or learned patterns must connect inputs to outputs transparently.

UNDER THE HOOD A decision process operationalizes the interface contract. Simplicity improves testability and explanation, but oversimplification can erase relevant distinctions. Learned components need explicit data, threshold, and fallback documentation.

Draw a fresh example with all three stages:

Every step connects a claim to observable evidence.

CASE A transparent sorter first checks item type, then one disambiguating feature, then sends unclear items to review.

STEP 1 — OBSERVE approved input fields

STEP 2 — TRACE smallest rule or learned decision flow with visible branches

STEP 3 — CONCLUDE suggested bin, not sure, and reason

MODEL REASONING Start with a baseline whose branches can be read aloud. Each input must affect a decision or be removed. Add not-sure and review paths, then explain when a learned component would earn its added complexity.

DO NOT OVERCLAIM Transparent does not mean correct; the flow still needs testing.

What new evidence could change this answer?

Concrete teaching case

A rule sends paper to P and everything else to M. Inputs include paper, metal and glass.

Trace glass and repair the rule.

Exact explanation to model

Glass incorrectly goes to M. Add an explicit metal check and a glass or review branch. Otherwise includes all remaining cases.

Explain the added worked example

A complete process says what happens for every kind of request in the task. It must handle a fitting activity, no fitting activity and unusable input. We will design a paper algorithm before building the prototype. Its decisions use the requirements from Week 30, so a different drawing should not silently change what counts as correct.

Keep the same catalogue order: B1 Bookmark drawing, 10 minutes, quiet Yes; B2 Story circle, 8 minutes, quiet No; B3 Pattern puzzle, 12 minutes, quiet Yes; B4 Tower challenge, 15 minutes, quiet No. These are fixed paper-model values with materials assumed available. Valid input is a whole number of minutes from 0 to 30 and Quiet required Yes or No.

A candidate fits when its required minutes are no more than the available minutes and either quiet is not required or the activity is quiet. Both parts must hold. The output is the first fitting activity in catalogue order, with a reason, or a clear no-match result. Invalid or missing input produces a clarification request. None of these outputs books materials or publishes a message.

Begin by checking the request. If minutes are missing, outside 0 to 30 or not a whole number, ask for a whole number in that range. If the quiet field is not Yes or No, ask for one of those values. State the fields needing correction and stop this run. When a corrected request arrives, validate it as a new run rather than continuing with old candidate state.

For a valid request, start at B1. Check time and the quiet condition for the current candidate. If both pass, output that candidate and stop. If either fails, move forward exactly one row. After B4 fails, there is no next row, so output no match and stop. Never jump back to B1 within the same search.

Key terms

Term · Meaning

Term: Validation

Meaning: Checking an input against the agreed rules before using it

Term: Branch

Meaning: A path chosen according to a condition

Term: Candidate

Meaning: An option currently being checked

Term: State

Meaning: Information that records the process's current position or values

Term: Termination

Meaning: Reaching a point where the process stops

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Validate · 9 is a whole number in range; No is valid · Current candidate B1

Validate: Check B1

9 is a whole number in range; No is valid: 10 minutes is more than 9

Current candidate B1: Move to B2

Validate: Check B2

9 is a whole number in range; No is valid: 8 is no more than 9; quiet is not required

Current candidate B1: Suggest B2 and stop

The current candidate changes from B1 to B2. The request values stay the same. After B2 passes, the process stops: it does not replace B2 with a later fitting row. This early stop preserves the first-match rule and gives the same required suggestion as the manual baseline, which marked every row before selecting the first fit.

Termination needs a reason. There are only four rows, each failed check moves to the next row, and no row is revisited. A valid request therefore examines at most four candidates. A success can stop sooner. Invalid input stops before the search. Merely drawing an arrow around a loop is not enough; the state must make progress toward an end.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Design the Decision Process

A fictional worksheet routes craft kits by required glue and number of scissors.

The facts we will use

K1: glue No, scissors 2. K2: glue Yes, scissors 2. K3: glue Yes, scissors 4.

Request: glue required Yes, scissors needed 2. Return first kit meeting both requirements.

Design the Decision Process

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Validate fields

Require Yes/No glue input and whole scissors count 0-4.



2. Evaluate K1

Scissors fit, but required glue is absent: continue.



3. Evaluate K2

Glue is present and 2 scissors meet the request: accept.



4. Stop

Return K2 without replacing it with later K3.

CHANGE ONE THING

Increase scissors needed to 3.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Validate fields

What happens and why: Require Yes/No glue input and whole scissors count 0–4.

Step: 2. Evaluate K1

What happens and why: Scissors fit, but required glue is absent: continue.

Step: 3. Evaluate K2

What happens and why: Glue is present and 2 scissors meet the request: accept.

Step: 4. Stop

What happens and why: Return K2 without replacing it with later K3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A complete decision process specifies what to do after each result. One passing requirement is insufficient. After a kit passes both, stopping preserves the first-match contract; automatically choosing the largest later kit would implement a different rule. A full scan with no accepted kit returns no match.

What this example does not establish

A correct flow depends on accurate catalogue entries and the declared requirements; the picture does not verify inventory.

Change one thing in the pictured case

Change: Increase scissors needed to 3.

Prediction: K3 is selected.

Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Demonstrate and guide practice

Board sequence

Write the given facts: A rule sends paper to P and everything else to M. Inputs include paper, metal and glass.

Write the question to explain: Trace glass and repair the rule.

Reveal after predictions: Glass incorrectly goes to M. Add an explicit metal check and a glass or review

branch. Otherwise includes all remaining cases.

Keep for an independent attempt: Write a decision process: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

Classroom dialogue

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks evidence; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

9 Yes: B1 takes 10, too long; B2 takes 8 but is not quiet; B3 takes 12, too long; B4 takes 15 and is not quiet. After all four fail, output no match. Rechecking a failed row without advancing could repeat forever with unchanged state. 31 No is outside the accepted time range, so request a whole time from 0 to 30 and stop before searching. Exit: B4 is compatible for 15 No, but violates the first-match rule; the required output is B1.

Case-specific teaching sequence

Stage · Teacher action

Stage: Session 1 start 0 to 7 minutes

Teacher action: Retrieve blank input versus a valid zero and no match.

Stage: 7 to 20 minutes

Teacher action: Identify all endings and branches using the diagram; complete W1.

Stage: 20 to 33 minutes

Teacher action: Trace 9 No and complete W2 with visible candidate state.

Stage: 33 to 45 minutes

Teacher action: Complete W3 and contrast both conditions with either condition.

Stage: Session 2 start 0 to 8 minutes

Teacher action: Retrieve the first-match rule and the meaning of No.

Stage: 8 to 23 minutes

Teacher action: Complete the guided no-match trace and demonstrate a loop without progress.

Stage: 23 to 38 minutes

Teacher action: Collect W4 independently; ask a partner to follow the written process literally.

Stage: 38 to 45 minutes

Teacher action: Complete W5 and the exit; save the versioned design for the prototype.

Reduce help deliberately

Completion task: Trace 9 minutes with Quiet required No. Record the current candidate, time result, quiet condition and next step until the process stops. Explain why the request values should not change as the candidate advances.

Teacher check: B1 fails time because 10 is greater than 9; quiet is unrestricted. Advance to B2. B2 passes time because 8 is no more than 9 and quiet is not required, so suggest B2 and stop. Candidate state changes to search the catalogue; changing the request values would solve a different task.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Designing a complete decision process. Proposed explanation: “One failed row proves no match”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Later rows may fit; continue until a fit or catalogue exhaustion.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Where does the first original rejection occur, and why is K3 not originally selected?

Reasoned teacher answer

K1 fails glue. K2 meets both conditions, so the first-match flow stops before considering K3.

Changed-case reasoning: K3 is selected. K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Collect each learner's explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the weekly evidence lab

Run the weekly evidence lab

Predict first, test honestly, and explain what the evidence means.

State your prediction before the result is known.

Complete the task: Create and trace the smallest useful decision flow.

Keep a normal case and an edge case clearly separated.

Record what happened—even when it does not match your prediction.

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing. Keep the expected result separate from an observed mistake.

Record · What to capture

Record: Input or evidence

What to capture: Exact data, instruction or claim

Record: Expected result

What to capture: Answer or safe fallback declared before the run

Record: Observed result

What to capture: Actual result, including errors

Record: Explanation

What to capture: Rule, calculation or evidence explaining the outcome

Record: Next test

What to capture: One justified change and a new prediction

Diagnose and reteach

CLAIM TO REPAIR Only successful labels need a branch.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation. Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

One failed row proves no match: Later rows may fit; continue until a fit or catalogue exhaustion.

Passing either condition is sufficient: The time and quiet conditions must both hold.

Any compatible answer satisfies the task: The first-in-catalogue tie rule is also a requirement.

A loop stops because it should: Show a state change that advances toward a finite end.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Assess and explain the answers

Worked case answer

Glass incorrectly goes to M. Add an explicit metal check and a glass or review branch. Otherwise includes all remaining cases.

Independent question

Write a decision process: a clear pencil goes to the pencil tray, a clear book to the shelf, and an unknown object to review. Trace a lunch box.

Expected independent answer

Check for a pencil, then a book; if neither is supported, request review. A lunch box follows the review branch. The fallback is part of the process.

What counts as evidence

Criterion · 0 points · 1 point · 2 points

Criterion: Result

0 points: Incorrect or absent

1 point: Partly correct

2 points: Correct with necessary conditions

Criterion: Reasoning

0 points: No supporting trace

1 point: Some steps

2 points: Clear mechanism, calculation or source support

Criterion: Limit

0 points: Overclaims certainty

1 point: Vague limitation

2 points: Relevant uncertainty or missing fact

Criterion: Transfer

0 points: Copies without adapting

1 point: Adapts with prompts

2 points: Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold. Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the evidence that challenges this claim: “Only successful labels need a branch.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Use if paper then P; else if metal then M; else review. Results are P, M, review, review. The catch-all must match the available actions. Never label all non-paper objects as metal merely because there are two trays.

Explanations for the additional workbook examples

W1 Describe all possible kinds of ending

Name the three kinds of final output in the activity process and the condition that leads to each. Explain

why a candidate failing its time check is not automatically a final no-match result.

Clarification follows invalid or missing input. A suggestion follows the first candidate passing both compatibility conditions. No match follows a valid request after all candidates fail. A single failed candidate does not settle whether later candidates fit, so the process must advance.

W2 Track the current candidate

Trace 9 minutes with Quiet required No. Record the current candidate, time result, quiet condition and next step until the process stops. Explain why the request values should not change as the candidate advances.

B1 fails time because 10 is greater than 9; quiet is unrestricted. Advance to B2. B2 passes time because 8 is no more than 9 and quiet is not required, so suggest B2 and stop. Candidate state changes to search the catalogue; changing the request values would solve a different task.

W3 Repair an incorrect branch

A design says “If the activity is quiet OR its time fits, suggest it.” For 5 minutes with Quiet required Yes it selects B1. Identify the failed requirement and replace the condition with a correct one. Give the correct result for this request.

B1 is quiet but requires 10 minutes, exceeding 5. Require time to fit AND the quiet condition to hold. The quiet condition is either quiet not required or a quiet activity. No catalogue activity fits 5 minutes, so the correct output is no match after checking all candidates.

W4 Design a complete room decision process

A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No. A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete process with validation, search, state change and stopping rules. Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

Validate group size and Yes or No; request clarification and stop if invalid. Start C1; check capacity and quiet. If both pass, suggest and stop; otherwise advance to C2. If C2 also fails, report no match and stop. For 6 No, C1 fails capacity, C2 passes capacity with quiet unrestricted, so suggest C2. Blank quiet requires clarification before searching. Each failed row advances, so at most two candidates are checked.

W5 Preserve the first compatible option

For the activity catalogue, 15 minutes with Quiet required No allows several activities. Explain why returning the last compatible row breaks the agreed rule even though it fits. Describe a stopping rule that preserves the first match, and one way to check that a loop cannot repeat forever.

All four activities fit, but the tie rule requires B1 first. Returning B4 meets compatibility but not the full specification. Stop immediately after the first passing candidate. Check that every failure advances exactly one row, no row is revisited and exhaustion of B4 ends with no match.

Assess independent transfer

Selected case: A new paper catalogue lists C1 first, capacity 4, quiet Yes; C2 second, capacity 8, quiet No. Valid requests give a whole group size from 1 to 10 and Quiet required Yes or No. A room fits if capacity is at least group size and the quiet condition is met; choose the first fit. Write a complete process with validation, search, state change and stopping rules. Trace 6 people No and explain the outcome for a blank quiet field. No booking takes place.

Selected case answer

Validate group size and Yes or No; request clarification and stop if invalid. Start C1; check capacity and quiet. If both pass, suggest and stop; otherwise advance to C2. If C2 also fails, report no match and stop. For 6 No, C1 fails capacity, C2 passes capacity with quiet unrestricted, so suggest C2. Blank quiet requires clarification before searching. Each failed row advances, so at most two candidates are checked.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect and extend learning

Connect and extend learning

Helpful Classroom Sorter

Complete the sorter v1 decision flow and annotate every branch with the evidence it uses.

Keep the decision, input, result, explanation and remaining limitation in the project evidence record.

Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two mechanisms with the same visible output and explain what evidence distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more evidence or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 32: Build a Paper Prototype. Retrieve today's evidence and identify what carries forward and what is new.

Support and fresh reassessment

Move a marker down the candidate cards while leaving the request card fixed. Use separate time and quiet ticks before combining them. Extension: compare the number of candidates examined by the early-stop process and the full manual checklist without claiming a measured time saving.

W4 must contain input validation, both room checks, first-match stopping, advancement, exhaustion and the C2 trace for 6 No. Blank quiet must produce clarification. After support change the request to 6 Yes: both rooms fail different conditions, so no match after two checks. Require an explanation that this is different from invalid input.

Week 32 builds a paper prototype from the decision process and records each step as a trace.

Plan the delayed return

Suggested return points in this level: ai-1/32, ai-1/34. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Draw the sorter's decision branches, loop-back arrows, final outputs and invalid-input path so every valid case can terminate.

7. Prepare, teach and assess

Teaching target: Design a decision process that covers valid, invalid and no-match requests

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: Trace 9 minutes with Quiet required No. Record the current candidate, time result, quiet condition and next step until the process stops. Explain why the request values should not change as the candidate advances.

Reasoned practice answer: B1 fails time because 10 is greater than 9; quiet is unrestricted. Advance to B2. B2 passes time because 8 is no more than 9 and quiet is not required, so suggest B2 and stop. Candidate state changes to search the catalogue; changing the request values would solve a different task.

Check every learner before discussion: Use the worked case for Designing a complete decision process. Proposed explanation: “One failed row proves no match”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Later rows may fit; continue until a fit or catalogue exhaustion.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Increase scissors needed to 3. Require a prediction and a reason before comparing. Expected result: K3 is selected. Reason: K1 lacks glue; K2 has only 2 scissors; K3 has glue and 4.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.